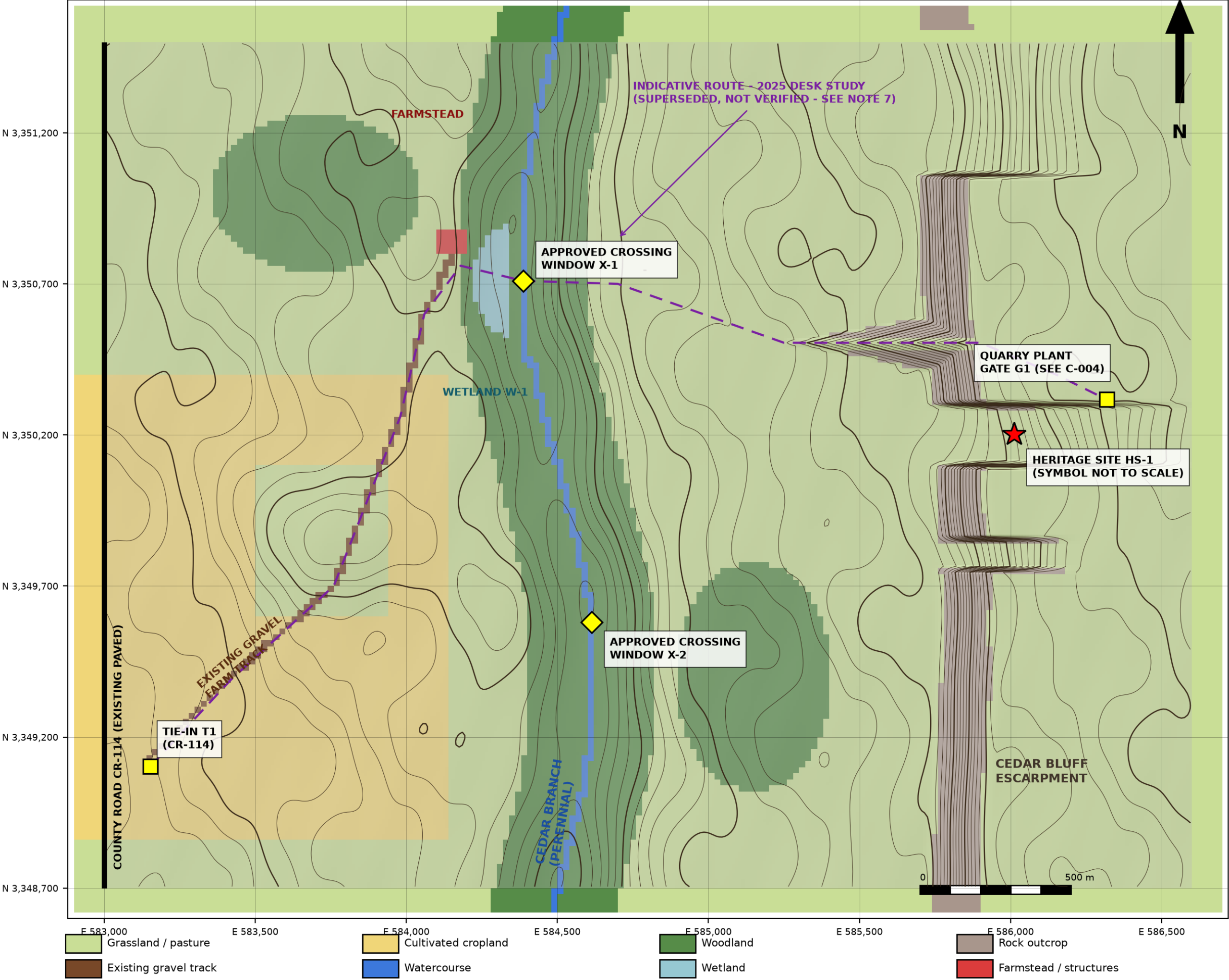


SITE CONSTRAINTS PLAN - PRESENTATION ONLY, DO NOT SCALE (SEE C-002 NOTE 1)



CLIENT
Cedar Bluff Aggregates LLC

PROJECT
Quarry Haul Road
Route Optioneering Study

DRAWING
**SITE CONSTRAINTS
PLAN**

REVISIONS

A 2026-08-03 Issued for route study

B 2026-09-14 Woodland clearing rate revised (C-002 Table 2)

C 2026-09-24 Plant gate G1 relocated; haul truck criteria and typical section (C-003)

D 2026-09-26 G1 re-surveyed; loaded haul, curve geometry and life-cycle cost basis added (Sheet C-004)

Rev D supersedes Rev C in full.

Horizontal: WGS 84 / UTM zone 14N (EPSG:32614), metres

Vertical: NAVD 88, US survey feet

NOT FOR CONSTRUCTION

SHEET
C-001

REV
D

Drawn: R.O. Checked: J.M. Date: 2026-09-26

DATA REGISTER, UNIT RATES AND ROUTING CONSTRAINTS

TABLE 1 - RASTER DATA REGISTER (issued with this drawing)

Item	CB-DEM-10m_heightmap_uint16.png	CB-LC-20m_landcover.png
Content	Bare-earth terrain model (LiDAR 2026-03)	Land-cover classification (field verified 2026-07)
Raster size	360 columns x 280 rows, 1 band	190 columns x 150 rows, 3 bands
Pixel type	16-bit unsigned integer (DN)	8-bit RGB, classes per Table 2
CRS	WGS 84 / UTM zone 14N (EPSG:32614)	WGS 84 / UTM zone 14N (EPSG:32614)
UL corner of UL pixel	E 583 000.00 m N 3 351 500.00 m	E 582 900.00 m N 3 351 620.00 m
Pixel size	10.00 m x 10.00 m (square)	20.00 m x 20.00 m (square)
Row / column order	Row 1 = north edge, column 1 = west edge	Row 1 = north edge, column 1 = west edge
Value	Elevation = 1150.00 + 0.01 x DN (US survey feet, NAVD 88)	Exact RGB triplet = class (no anti-aliasing)

TABLE 2 - LAND-COVER CLASSES AND BASE UNIT RATES (REV B, UNCHANGED AT REV C AND REV D)

Code	Class	RGB	Base unit rate c (USD per metre of road)
1	Grassland / pasture	(201, 222, 150)	420
2	Cultivated cropland	(240, 214, 120)	480
3	Woodland	(86, 140, 72)	940 (Rev B; Rev A value 690 withdrawn)
4	Rock outcrop	(168, 150, 140)	1,150
5	Existing gravel track	(120, 72, 40)	160
6	Watercourse	(60, 120, 220)	PROHIBITED (Note 3)
7	Wetland	(150, 200, 210)	PROHIBITED
8	Farmstead / structures	(220, 60, 60)	PROHIBITED
-	Culvert at approved crossing window	-	3,800 (Note 3)

TABLE 3 - CONTROL POINTS (WGS 84 geographic, decimal degrees)

Point	Latitude	Longitude
T1 Tie-in to CR-114 (route start)	30.2707418	-98.1355013
G1 Quarry plant gate (route end) - REV C VALUE, SUPERSEDED: SEE C-004 TABLE 6	30.2814772	-98.1024981
X-1 Approved crossing window centre	30.2851604	-98.1225516
X-2 Approved crossing window centre	30.2749484	-98.1202827
HS-1 Heritage site (recorded point)	30.2804614	-98.1056944

NOTES

1. The two rasters in Table 1 govern. Sheet C-001 is a presentation plot of them and is not to be scaled or digitised.
2. Route start and end are the DEM cells containing control points T1 and G1.
3. Watercourse cells may be traversed only at approved crossing windows X-1 and X-2: a watercourse cell is passable where its DEM cell centre lies within 30.0 m of a window centre, and is then costed at the culvert rate. All other watercourse cells are prohibited.
4. Wetland W-1: wetland cells are prohibited and no part of the road may lie within 30.0 m of the mapped wetland boundary.
5. Heritage site HS-1: exclusion zone of radius 150.0 m about the recorded point (Texas Historical Commission consultation, 2026).
6. Maximum grade 10.0 percent for loaded 40 t haul trucks, in either direction of travel.
7. The indicative route on C-001 is from a 2025 desk study that pre-dates the LiDAR, the land-cover survey and Notes 3 to 5.
8. Haul truck criteria: Sheets C-003 and C-004. Precedence where sheets differ: C-004, then C-003, then this sheet.

CLIENT
Cedar Bluff Aggregates LLC

PROJECT
Quarry Haul Road
Route Optioneering Study

DRAWING
DATA REGISTER,
UNIT RATES AND
CONSTRAINTS

- REVISIONS
- A 2026-08-03 Issued for route study
- B 2026-09-14 Woodland clearing rate revised (C-002 Table 2)
- C 2026-09-24 Plant gate G1 relocated; haul truck criteria and typical section (C-003)
- D 2026-09-26 G1 re-surveyed; loaded haul, curve geometry and life-cycle cost basis added (Sheet C-004)

Rev D supersedes Rev C in full.

Horizontal: WGS 84 / UTM zone 14N (EPSG:32614), metres

Vertical: NAVD 88, US survey feet

NOT FOR CONSTRUCTION

SHEET
C-002

REV
D

Drawn: R.O. Checked: J.M. Date: 2026-09-26

HAUL ROAD TYPICAL SECTION AND HAUL TRUCK OPERATING CRITERIA

TYPICAL SECTION A-A (TANGENT, CUT/FILL NOT SHOWN) SCALE 1:100 AT A3 - DO NOT SCALE FROM THIS PLOT
CL (route centreline = DEM cell centres, C-002 Note 2)

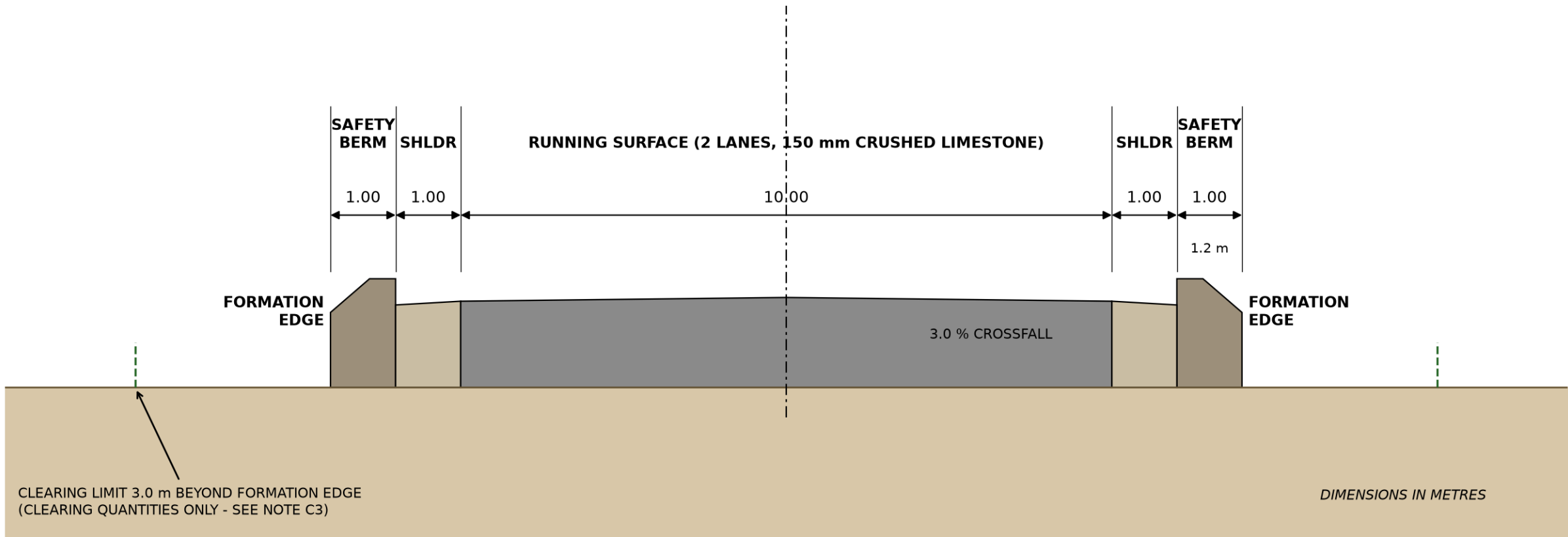


TABLE 4 - HAUL TRUCK OPERATING CRITERIA (40 t ARTICULATED HAUL TRUCK, LOADED)

Ref	Criterion	Value	Application to the route study
T1	Maximum grade, either direction	10.0 %	Every move; as C-002 Note 6
T2	Sustained steep grade	> 8.0 %	A move steeper than 8.0 % is a steep move
T3	Maximum continuous run of steep moves	60.0 m	Horizontal length; any move at 8.0 % or flatter ends the run
T4	Maximum change of direction between consecutive moves	45 deg	Measured between consecutive grid moves; the first move from the start cell is unrestricted
T5	Design speed (loaded)	40 km/h	Information only
T6	Rated payload / gross vehicle mass	40 t / 72 t	Information only

NOTES

- C1. 'The road' in C-002 Notes 4 and 5 means the full formation shown in Section A-A. Wetland setback and heritage exclusion are measured to the formation edge on either side of the centreline, along every straight move between cell centres.
- C2. The crossing-window test of C-002 Note 3 is applied to centreline cell centres only.
- C3. The clearing limit and batters are for quantities only and are not routing constraints.
- C4. Criteria T1 to T4 apply together. Unit rates (C-002 Table 2) and the grade factor are unchanged by this sheet.

CLIENT
Cedar Bluff Aggregates LLC

PROJECT
Quarry Haul Road
Route Optioneering Study

DRAWING
TYPICAL SECTION
AND HAUL TRUCK
CRITERIA

- REVISIONS
- A 2026-08-03 Issued for route study
 - B 2026-09-14 Woodland clearing rate revised (C-002 Table 2)
 - C 2026-09-24 Plant gate G1 relocated; haul truck criteria and typical section (C-003)
 - D 2026-09-26 G1 re-surveyed; loaded haul, curve geometry and life-cycle cost basis added (Sheet C-004)

Rev D supersedes Rev C in full.

Horizontal: WGS 84 / UTM zone 14N (EPSG:32614), metres
Vertical: NAVD 88, US survey feet

NOT FOR CONSTRUCTION

SHEET
C-003

REV
D

Drawn: R.O. Checked: J.M. Date: 2026-09-26

LOADED HAUL, ROUTE GEOMETRY AND LIFE-CYCLE COST BASIS (REV D)

TABLE 5 - ADDITIONAL ROUTE SCREENING CRITERIA (APPLY WITH C-003 TABLE 4)

Ref	Criterion	Value	Definition / application
D1	Direction of loaded travel	G1 to T1	Product leaves the plant gate loaded and is hauled to CR-114; trucks return empty
D2	Rolling resistance, running surface	3.0 %	Section A-A crushed limestone surface - applies to every move of the route
D3	Max. effective grade, loaded uphill	9.0 %	Effective grade = grade resistance + rolling resistance, loaded direction (D1)
D4	Max. grade, loaded downhill / empty	T1	C-003 Table 4 T1 (10.0 %) - unchanged
D5	Min. rate of vertical curvature K	1.4 m/%	K = L / A (m per 1 % grade change) for every pair of consecutive moves; A = algebraic difference of the two grades (%); L = distance along the centreline between the two move midpoints
D6	Min. horizontal curve radius R	45.0 m	A circular arc of radius R is fitted at every deflection vertex, tangent to both moves; arcs may not overlap and may not extend past the start or end cell centre (Note D3)
D7	Loaded haulage cost (PV, 20-year life)	8,000 /m	USD per metre of vertical rise climbed by loaded trucks (D1), charged only on moves whose effective grade (as D3) exceeds 4.0 % (top-gear range); falls are not credited

TABLE 6 - CONTROL POINT G1, RE-SURVEYED 2026-09-25 (SUPERSEDES C-002 TABLE 3 ROW G1)

Point	CRS / coordinate type	Easting	Northing
G1 Quarry plant gate (route end)	NAD83 / Texas Central (ftUS), EPSG:2277 - TxDOT SURFACE	3,001,506.17 ft	10,073,679.76 ft
	Surface adjustment factor SAF = 1.00012, scaled about grid origin (0, 0): grid = surface / SAF		

TABLE 7 - REFERENCE VALUES (INFORMATION ONLY - NOT ROUTE CRITERIA)

Item	Value	Source
Rolling resistance, in-pit ramps (not this road)	2.0 %	Mine plan 2025
Rolling resistance, unsurfaced haul road (not permitted)	6.0 %	Mine plan 2025
Haul truck minimum turning radius (outer front wheel)	8.9 m	Manufacturer data
Haulage cost per metre of fall, loaded (brake wear, info)	450 USD	Operator estimate - not in life-cycle basis
Top-gear effective grade limit, empty truck	6.0 %	Manufacturer data

NOTES

- D1. The route to be selected minimises the TOTAL route cost = construction cost (C-002 Table 2 rates x grade factor, unchanged) + loaded haulage cost (Table 5 D7). Report both components and the total.
- D2. Grades on long sections and in vertex schedules are signed in the direction of increasing chainage (T1 to G1). Criteria D3 and D7 are defined in the loaded direction (D1), which is opposite to increasing chainage.
- D3. A deflection vertex is an interior route vertex at which the direction of travel changes; on the 10 m grid each deflection is 45 deg (C-003 T4). The start and end cell centres are not deflection vertices. The arcs are a screening check only: costs, grades and clearances stay on the straight moves between cell centres.
- D4. D5 applies to every pair of consecutive moves, including pairs meeting at a deflection vertex. It does not apply before the first move.
- D5. Table 6 replaces the G1 latitude/longitude on C-002. C-001 still shows G1 at its Rev C position (presentation only).
- D6. Where this sheet differs from C-002 or C-003 this sheet governs. C-003 Table 4 T1 to T4 remain in force.

CLIENT
 Cedar Bluff Aggregates LLC

PROJECT
 Quarry Haul Road
 Route Optioneering Study

DRAWING
 LOADED HAUL,
 GEOMETRY AND
 COST BASIS

- REVISIONS
- A 2026-08-03 Issued for route study
 - B 2026-09-14 Woodland clearing rate revised (C-002 Table 2)
 - C 2026-09-24 Plant gate G1 relocated; haul truck criteria and typical section (C-003)
 - D 2026-09-26 G1 re-surveyed; loaded haul, curve geometry and life-cycle cost basis added (Sheet C-004)

Rev D supersedes Rev C in full.

Horizontal: WGS 84 / UTM zone 14N (EPSG:32614), metres

Vertical: NAVD 88, US survey feet

NOT FOR CONSTRUCTION

SHEET
 C-004
 REV
 D

Drawn: R.O. Checked: J.M. Date: 2026-09-26